Gavin D. Milligan

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EDUCATION

University of California, Irvine *B.S. in Software Engineering* **Class of 2026**

EXPERIENCE & LEADERSHIP

American Express **Phoenix, AZ**

Software Engineer I *March 2026 - Present*

- Engineered an enterprise GenAI platform with full-stack Python and React.js, leveraging pgvector, OpenAI text embeddings, and RAG to automate high-volume merchant pricing and display valuable business insights.
- Led a large-scale architecture migration, consolidating **14 separate** REST API repositories into a single, highly optimized monorepo responsible for managing the data which processes over **$40 billion** in transactions yearly.
- Drove the technical direction and UI design among a team of **10 engineers**, successfully pitching winning design architectures and shipping scalable, secure features using React, JavaScript, Java, and Github actions

NinjaTrader **Chicago, IL**

Software Development Engineer in Test Intern *May 2025 - July 2025*

- Created an automated workflow using AI agents and Zephyr to generate test cases for each new pull request company-wide, reducing QA effort from **1 hour** to less than **2 minutes**, while keeping compute costs low
- Developed JavaScript modules leveraging Playwright and an MCP server to parse DOM structures, create reusable maps, and support scalable automated test execution used by **10+** engineers
- Built robust API endpoints for internal tools using an MVC architecture (Express.js) for the DevOps and QA teams

Myfye **Irvine, CA**

Technical Co-Founder, Full Stack *February 2024 - September 2025*

- Built a secure financial application, focused on blockchain-abstracted asset transfers, enabling tokenized representations of U.S. Treasuries and equities; won **$10k** at the Solana Colosseum Hackathon
- Developed sponsored transactions by creating a transaction object, sending it to a secure backend for signing, and then back to the frontend to be signed again by the user's embedded wallet, eliminating fees for users
- Optimized transactional processing latency by engineering a high-performance system for network data exchange, achieving an average **0.8-second**, independently verifiable and final data commit on a distributed ledger
- Built a deposit and withdrawal service using PayPal, Privy and Coinbase Onramp

Blockchain At UCI **Irvine, CA**

Technical Lead *June 2023 - May 2025*

- Engineered the architecture for a secure rental marketplace, securing a victory at the ETHDenver '25 hackathon
- Managed open source GitHub repositories by assigning tasks, approving pull requests, and overseeing branch management for a team of **12** developers, after conducting over **30** technical interviews and selecting the best

Coder School **Irvine, CA**

Code Coach *August 2024 - January 2025*

- Taught *Python*, *C++, and HTML/CSS* to students in grades 1-10, creating lesson plans to match their skill levels and providing 1:1 support to address challenges and improve understanding

PROJECTS

FinFriend | *Swift, Kotlin, React, Bitcoin, Cybersecurity* *October 2022 - August 2023*

- Built a create-your-own-currency platform for iOS, Android, and the web that has made over **$1k in revenue** and has seen over **100 currencies** launched

SKILLS & TECHNOLOGIES

Languages: Python, Java, C++, JavaScript, HTML, CSS, Swift, Kotlin, SQL

Tools / Frameworks: AWS, Google Cloud, REST APIs, React, Redux, Git, Apache, Kafka, Node.JS, RAG, Vector DBs, Artificial Intelligence, Bash, Docker, UML Diagramming, Object-Oriented Design, Software Development Lifecycle, iOS, Android, Unit Testing / Integration Testing, Quality Assurance, Automation, Jira, Confluence, Scrum, Agile, Postgres